Bobby Tromm

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Education

2025–Present **PhD, Computational Neuroscience**, *University College London*
Supervisors: Caswell Barry, Tim Behrens
Thesis: Cognitive maps in biological and artificial neural networks

2021–2023 **MSc, Cognitive and Clinical Neuroscience**, *Maastricht University*

2016–2020 **BS, Neuroscience**, *Brandeis University*

Research

2024–2025 **Research Engineer**, *Neuronal Circuits & Brain Dynamics, Paris Brain Institute*
Supervisor: Nikolas Karalis

2023 **Visiting Researcher**, *Department of Psychiatry, University of Oxford*
Supervisor: Morten Kringelbach

2022–2023 **Research Assistant**, *Department of Psychopharmacology, Maastricht University*
Supervisor: Jan Ramaekers

2019–2020 **Research Assistant**, *Volen National Center for Complex Systems, Brandeis University*
Supervisor: Eve Marder

Publications

2025 C. Coleman, K. Shinozuka, **R. Tromm** *et al.* The role of the dorsolateral prefrontal cortex in ego dissolution and emotional arousal during the psychedelic state. *Human Brain Mapping.*

Conferences

2023–2024 **R. Tromm**, C. Timmermann, P. Mallaroni, N. Mason, R.L. Carhart-Harris, G. Deco, J. Ramaekers, and M. Kringelbach. *The anarchically organized brain: changes in functional hierarchy in altered states.* Poster presented at ASSC 27, Tokyo (2024) and ALPS 2023, University of Geneva (2023).

2019 **R. Tromm**, C. Guerini, S. Gorur-Shandilya, E. Marder. *Dual homeostatic mechanisms can reproduce diverse ion channel correlations.* SciFest IX. Brandeis University. (Poster)

Teaching

2026–Present **BSc Project Supervisor**, *University College London*
Necessary architectures for grid cells

2025–Present **Postgraduate Teaching Assistant**, *University College London*
- COMP0085 Approximate Inference and Learning in Probabilistic Models (2026)
- COMP0086 Probabilistic and Unsupervised Learning (2026)
- NEUR0016 Neural Computation (2025)

Selected Honors & Awards

2025 **Research Excellence Scholarship**, *University College London*
£175,000, awarded to the top 0.5% of PhD applicants

2024 **Polaris Fellowship**, *Entrepreneur First*
Fellowship for technical researchers pursuing commercial research projects; £3,000 grant

2024 **Usona Institute Scholarship**, *Usona Institute*
Awarded twice (2023, 2024), $7,500 each, for academic and research achievements in psychedelic studies

2020 **Justice Brandeis Scholarship**, *Brandeis University*
$140,000, offered to top applicants

Service

2026–Present **Organiser, AI Workshop 2027**, *Mediterranean Society for Consciousness Science*
Convening a workshop bridging consciousness science and mechanistic interpretability

2026–Present **As1 Next Generation Advisory Council**, *Burroughs Wellcome Fund*

2024–Present **Student Committee**, *Association for the Scientific Study of Consciousness*

2019–2021 **Research and Professional Development Committee**, *Intercollegiate Psychedelics Network*

Additional Training

2026 **School on Analytical Connectionism**, *Chalmers University of Technology*
Statistical physics of learning and mechanistic interpretability of language acquisition

2026 **Deep Learning**, *Neuromatch Academy*
Selective summer school in deep learning

2025 **Gatsby Bridging Programme**, *Gatsby Computational Neuroscience Unit, UCL*
Mathematics summer school: linear algebra, calculus, probability, differential equations, Fourier analysis

2024 **Neurohackademy**, *University of Washington*
Summer school in neuroimaging and data science

2024 **QStr Summer School**, *ATR; Monash University*
Summer school on category theory and information theory in cognitive science

2020 **Computational Neuroscience**, *Neuromatch Academy*
Single-neuron, dynamical systems, and stochastic modelling